CSE330

Chapter 4 : Root finding for non linear equations

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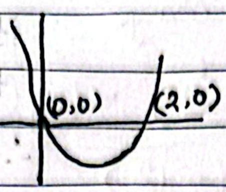
4.1 Bisection Method

$f(x) = x^2 - 2x = 0$

$\Rightarrow x(x-2) = 0$

$\Rightarrow x = 0, 2$

Roots $\rightarrow x^* = 0, 2$



- Non Linears

  # Polynomial degree ≥ 2

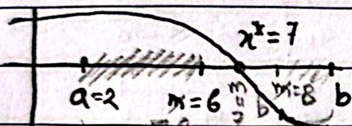
  # Trigonometric function (sin, cos)

  # Exponential function (e^x)

  # $(\frac{1}{x} - 2)$ $\boxed{\text{Fractional}}$

$\Rightarrow$ Bisection Root Finding Algorithm (Interval Bisection method):

$f(x) = - - -$, $[a, b]$ . . - find x^*



$m = \frac{a+b}{2} = \frac{2+10}{2} = 6$

$[a, m]$ $[m, b]$ a=6. b=10

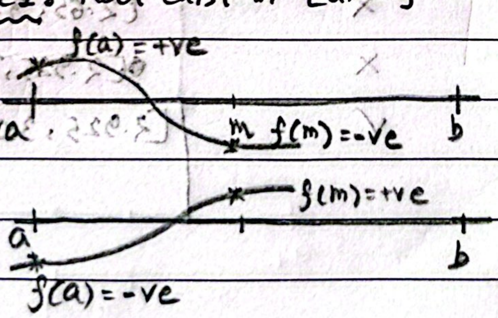
$[a, m]$ $[m, b]$ a=6. b=8

$m = \frac{6+8}{2} = 7$

Binary Search

$f(m) = f(7) = 0$; m=7 is the root

Case1: Root exist in $[a, m]$

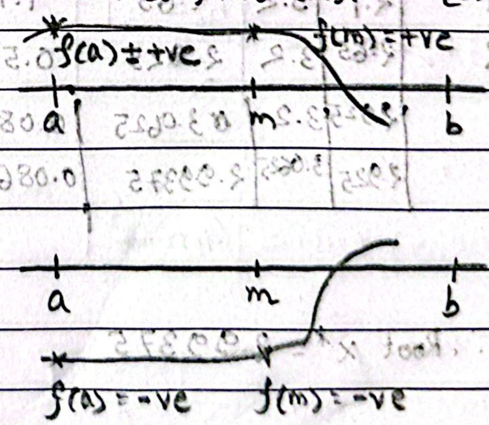


f(a) & f(m) has different sign.

✱ if $f(a) * f(m) = -ve$

then Root exist in $[a, m]$, $b = m$

Case 2: Root does not exist in $[a, m]$



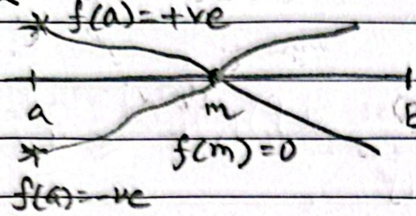
f(a) and f(m) have similar sign.

✱ $f(a) * f(m) = +ve$

then Root doesn't exist in $[a, m]$

$a = m$

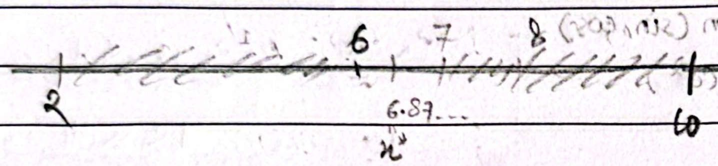
Case 3:



if $f(a) * f(m) = 0$

then $f(m) = 0$

$\therefore$ m is the root.



There is a limit for iteration. We do this until we find actual roots or we reach machine epsilon ε

↳ Stopping Criteria

Using Bisection method, find the root of $f(x) = x^3 - 7x^2 + 14x - 6$ in interval $[1, 3.2]$. Your solution must be accurate within ~~10~~ 0.05.

Interval	a	b	$m = \frac{a+b}{2}$	f(a)	f(m)	Root exist in[a,m]	new interval
0	1	3.2	2.1	2	1.79	×	[m,b]=[2.1,3.2]
1	2.1	3.2	2.65	1.79	0.55	×	[2.65, 3.2]
2	2.65	3.2	2.925	0.55	0.086	×	[2.925, 3.2]
3	2.925	3.2	3.0625	0.086	-0.054	✓	[2.925, 3.0625]
4	2.925	3.0625	$\underbrace{2.99375}_{\text{root}}$	0.086	$\underbrace{0.006}_{<\varepsilon}$		

$\therefore$ Root $x^* = 2.99375$

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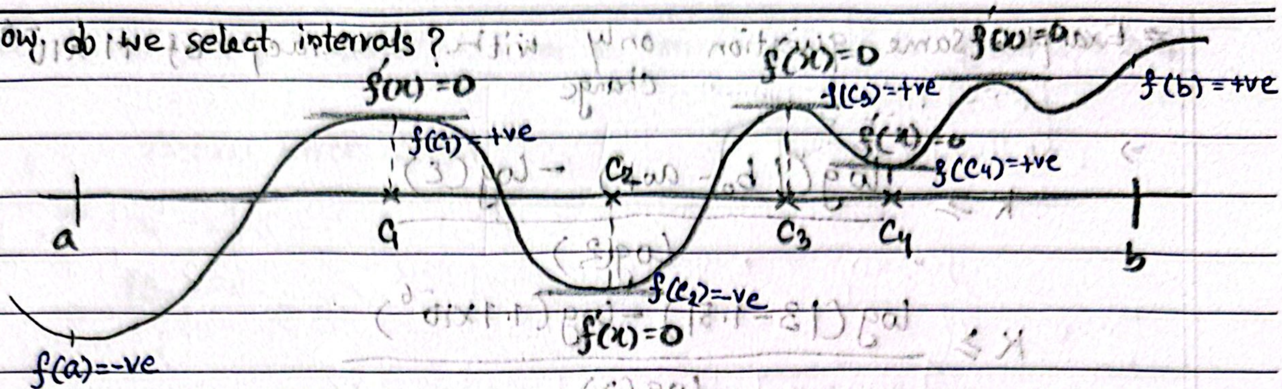
How do we select intervals?



$$[a,b] = [a,c_1] \cup [c_1,c_2] \cup [c_2,c_3] \cup [c_3,c_4] \cup [c_4,b]$$

$[a,c_1]$ $[c_1,c_2]$ $[c_2,c_3]$

How many iteration is required for given intervals & epsilon value to reach the root?

$\Rightarrow$ $[a,b]$, ε

Number of iteration Required, $K \geq \dfrac{\log(|b_0 - a_0|) - \log(\varepsilon)}{\log(2)} \boxed{-1}$

Don't include if Accuracy is specified

Example: Find the number of iteration required to get the root with error bound of machine epsilon, $\varepsilon = 1.1 \times 10^{-6}$ in interval $[1.5, 3]$

$\Rightarrow$

$$K \geq \frac{\log(|b_0 - a_0|) - \log(\varepsilon)}{\log(2)} - 1$$

$$K \geq \frac{\log(|3 - 1.5|) - \log(1.1 \times 10^{-6})}{\log(2)} - 1$$

$K \geq 19.379$

$K \geq 20$

$\therefore$ number of iteration require is 20.

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Example: Same question only change with accuracy of 1.1×10^{-6}

$\Rightarrow$

$$K \geq \frac{\log(|b_0 - a_0|) - \log(\varepsilon)}{\log(2)}$$

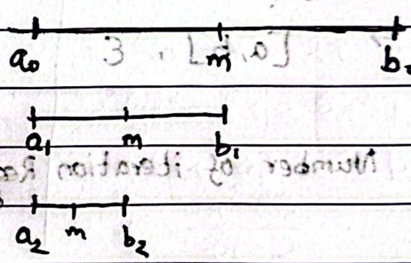
$$K \geq \frac{\log(|3-1.5|) - \log(1.1 \times 10^{-6})}{\log(2)}$$

$K \geq 20.379$

$K \geq 21$

⊛ Bisection number of iteration formula derivation:

For interval, a_0, b_0:



$$|b_1 - a_1| = \frac{|b_0 - a_0|}{2}$$

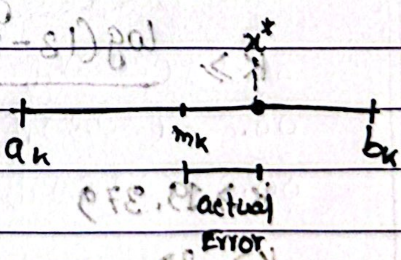
$$|b_2 - a_2| = \frac{|b_1 - a_1|}{2} = \frac{|b_0 - a_0|}{2^2}$$

$$|b_3 - a_3| = \frac{|b_0 - a_0|}{2^3}$$

$\vdots$

$$|b_k - a_k| = \frac{|b_0 - a_0|}{2^k} \quad \text{———} (1)$$

After k iteration,



Actual error $= |m_k - x^*|$

Intuitively,

Actual Error $\leq \frac{|b_k - a_k|}{2}$

$\Rightarrow |m_k - x^*| \leq \frac{|b_k - a_k|}{2}$

$\Rightarrow |m_k - x^*| \leq \frac{|b_0 - a_0|}{2^{k+1}}$ [From eqn 1]

For maximum possible error,

$$|m_k - x^*|_{max} = \frac{|b_0 - a_0|}{2^{k+1}} \quad \text{———} (2)$$

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Since, we stop our iteration when actual error is less than given epsilon,

Actual Error $\leq \varepsilon$

$\Rightarrow |m_k - x^*| \leq \varepsilon$

$\Rightarrow \dfrac{|b_0 - a_0|}{2^{k+1}} \leq \varepsilon$ [From eq^n 2]

$\Rightarrow \dfrac{|b_0 - a_0|}{\varepsilon} \leq 2^{k+1}$

$\Rightarrow \log\left(\dfrac{|b_0 - a_0|}{\varepsilon}\right) \leq \log(2^{k+1})$

$\Rightarrow \log(|b_0 - a_0|) - \log(\varepsilon) \leq (k+1)\log(2)$

$\Rightarrow k+1 \geq \dfrac{\log(|b_0 - a_0|) - \log(\varepsilon)}{\log(2)}$

$\therefore k \geq \dfrac{\log(|b_0 - a_0|) - \log(\varepsilon)}{\log(2)} - 1$

Disadvantage of Bisection method:

1. Slow convergence
2. Can't find multiple ~~loops~~ roots in single interval.

Advantage

1. Even though it is slow it is more robust and guaranteed.

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4.2 : Fixed Point Iteration:

- Root Finding Algorithm
- Iterative Method

$f(x)=0$ → We try to manipulate this function

Mathematically Manipulation

$\boxed{g(x)=x}$ ← $\boxed{\text{Condition}}$

→ x is the root of $f(x)$

Step 1. Finding $g(x)$:

Given $f(x)=x^2-2x-3$. Construct 3 $g(x)$ from $f(x)$

$f(x)=0$

$\Rightarrow x^2-2x-3=0$	$x^2-2x-3=0$	$x^2-2x-3=0$
$\Rightarrow x^2=2x+3$	$\Rightarrow x^2-x-x-3=0$	$\Rightarrow 2x^2-x^2-2x-3=0$
$\Rightarrow \underbrace{x}_{x}=\underbrace{\sqrt{2x+3}}_{g(x)}$	$\Rightarrow x^2-x-3=x$	$\Rightarrow 2x^2-2x=x^2+3$
	$\Rightarrow x=x^2-x-3$	$\Rightarrow x(2x-2)=x^2+3$
$\therefore g_1(x)=\sqrt{2x+3}$	$\therefore g_2(x)=x^2-x-3$	$\Rightarrow x=\dfrac{x^2+3}{2x-2}$
		$\therefore g_3(x)=\dfrac{x^2+3}{2x-2}$

Step 2. On a given initial point $x_0(=0)$; Find the root iteratively:

Find the root using fixed point iteration for ~~x^2-2x-3~~ $f(x)=x^2-2x-3$ using the initial point $x_0=0$. [Use upto 3 significant figure.]

$f(x)=0$

$\Rightarrow x^2-2x-3=0$

$\Rightarrow x=\sqrt{2x+3}$

$g_1(x)=\sqrt{2x+3}$

Now, $x_0=0$ $\quad g_1(x_0)=g_1(0)=1.73$

Iteration 1: $g_1(0)=1.73$

" 2: $g_1(1.73)=2.54$

" 3: $g_1(2.54)=2.84$

" 4: $g_1(2.84)=2.95$

Converge [we reached the root.]

Iteration 5: $g_1(2.95)=2.98$

" 6: $g_1(2.98)=3.00$

" 7: $g_1(\underbrace{3.00}_{x})=\underbrace{3.00}_{x}$ $\quad \boxed{g_1(x)=x}$ [Fixed point Reach]

$\therefore$ Root, $x^*=3$

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$g_2(x) = x^2 - x - 3$

$x_0 = 0$

$g_2(0) = -3.00$

$g_2(-3) = 9.00$

$g_2(9) = 69.00$

$g_2(69) = 4.69 \times 10^3$

Divergence
(As we did not reach the root)

$g_3(x) = \dfrac{x^2+3}{2x-2}$

$x_0 = 0$

$g_3(0) = -1.50$

$g_3(-1.50) = -1.05$

$g_3(-1.05) = -1.00$

$g_3(-1.00) = -1.00$

Convergence
[We reached the root]

$\therefore g_3(x) = x$

$\therefore$ Root, $x^* = -1$

Actual Root: $x^2 - 2x - 3 = 0$

$\Rightarrow x^2 - 3x + x - 3 = 0$

$\Rightarrow x(x-3) + 1(x-3) = 0$

$\Rightarrow (x-3)(x+1) = 0$

$\therefore x = -1, 3$

Let, $x_0 = 42$,

$g_3(x) = \dfrac{x^2+3}{2x-2}$

$g_3(42) = 21.6$

$g_3(21.6) = 11.4$

$g_3(11.4) = 6.39$

$g_3(6.39) = 4.07$

$g_3(4.07) = 3.19$

$g_3(3.19) = 3.01$

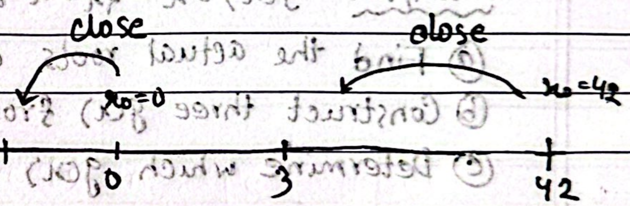
~~$g_3(3.01) =$~~

$g_3(3.01) = 3.00$

$g_3(3.00) = 3.00$

$g_3(x) = x$

$\therefore x^* = 3$



The root we converge to depends on the choice of the initial point (x_0) and the chosen $g(x)$.

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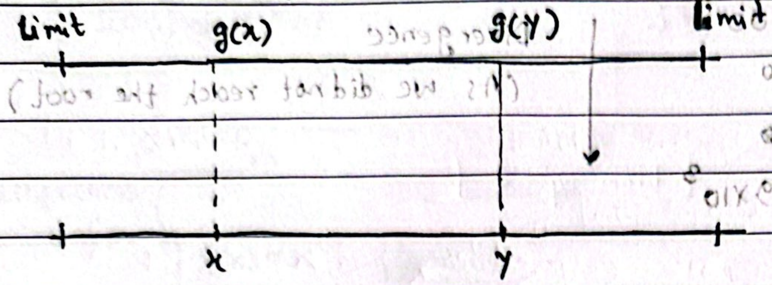
How do we know which $g(x)$ is convergent?

=> **Contraction Mapping Theorem**



$$\lambda = \left| \frac{g(y) - g(x)}{y - x} \right|$$

Gradient of $g(x)$ (slope)

$\therefore$ Converging Rate, $\lambda = |g'(x)|$

1. $\rightarrow \lambda = 0 \longrightarrow$ Super Linear Convergent.
   [Less number of iteration to find the root]
2. $\rightarrow 0 < \lambda < 1 \longrightarrow$ Linear Convergent
   [more iteration is required]
3. $\rightarrow \lambda \geq 1 \longrightarrow$ Divergent.
   [Will not find the root]

Example: $f(x) = x^3 - 2x^2 - x + 2$

ⓐ Find the actual roots of this equation.

ⓑ Construct three $g(x)$ from $f(x)$.

ⓒ Determine which $g(x)$ are convergent & which are divergent?

OR, Find the converging $g(x)$ and which root it will converge to?

=> ⓐ $x^3 - 2x^2 - x + 2 = 0$

$\Rightarrow x^2(x-2) - 1(x-2) = 0$

$\Rightarrow (x^2-1)(x-2) = 0$

$\Rightarrow (x-1)(x+1)(x-2) = 0$

$\therefore x = 1, -1, 2$

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(b)

$x^3-2x^2-x+2=0$	$x^3-2x^2-x+2=0$	$x^3-2x^2-x+2=0$
$\Rightarrow 2x^2 = x^3-x+2$	$\Rightarrow x = x^3-2x^2+2$	$\Rightarrow x(x^2-2x-1)+2=0$
$\Rightarrow x^2 = \frac{1}{2}(x^3-x+2)$	$\therefore g_2(x) = x^3-2x^2+2$	$\Rightarrow x = \frac{-2}{x^2-2x-1}$
$\Rightarrow x = \sqrt{\frac{1}{2}(x^3-x+2)}$		$\therefore g_3(x) = \frac{-2}{x^2-2x-1}$
$\therefore g_1(x) = \sqrt{\frac{1}{2}(x^3-x+2)}$		

$\therefore g_1(x) = \frac{1}{\sqrt{2}}(x^3-x+2)^{1/2}$

(c) $\lambda_1 = |g_1'(x)| = \left|\frac{d}{dx}\frac{1}{\sqrt{2}}(x^3-x+2)^{1/2}\right|$

$= \left|\frac{1}{\sqrt{2}}\cdot\frac{1}{2}(x^3-x+2)(3x^2-1)\right|$

From (a), $x=-1, 1, 2$, For, $x=-1$, $\lambda_1=0.6$ Linear Convergent

$x=1$, $\lambda_1=0.5$ Linear Convergent

$x=2$, $\lambda_1=3.75$ Divergent

$\lambda_2 = |g_2'(x)| = \left|\frac{d}{dx}(x^3-2x+2)\right|$

$= |3x^2-2|$

For $x=-1$, $\lambda_2=1$ Divergent

$x=1$, $\lambda_2=7$ Divergent

$x=2$, $\lambda_2=4$ Divergent

$\lambda_3 = |g_3'(x)| = \left|\frac{d}{dx}\left(\frac{-2}{x^2-2x-1}\right)\right|$

$= |-2(-1)(x^2-2x-1)^{-2}(2x-2)|$

$= \left|2\frac{2x-2}{(x^2-2-1)^2}\right|$

For $x=-1$, $\lambda_3=0$ Super Linear Convergent

$x=1$, $\lambda_3=2$ Divergent.

$x=2$, $\lambda_3=4$ Divergent.

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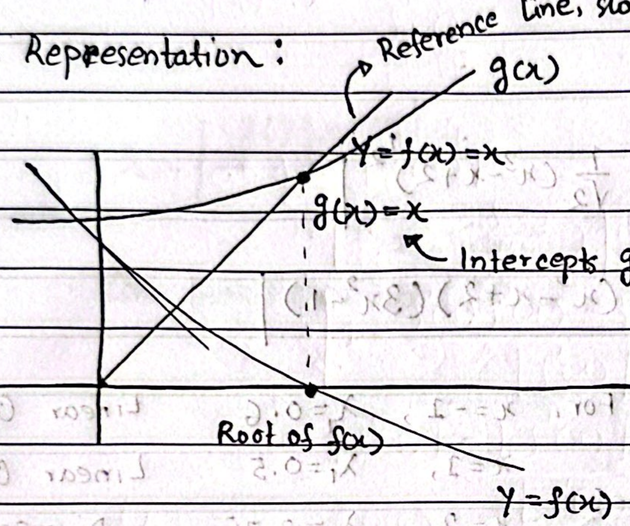
Advantage

1. Convergence very fast if $\lambda = 0$ or $g(x)$ is super linear convergent

Disadvantage

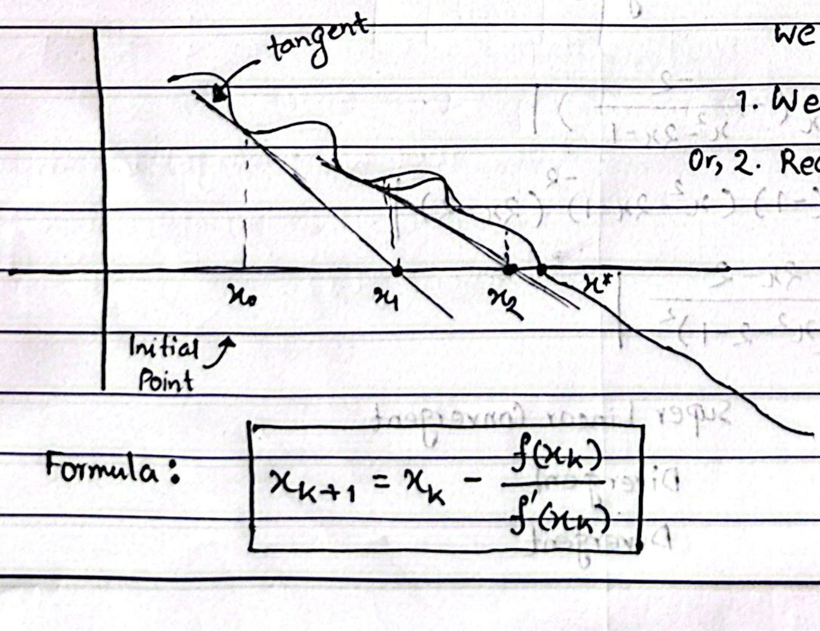
1. Not always guaranteed – as it depends on $g'(x)$ & initial point

* Graph Representation:



4.4: Newton's Method / Newton Raphson Method:

- Root Finding Algorith
- Super linear convergent ($\lambda = 0$)
- Iterative Approach



We keep iterating untill
1. We are reaching the root
Or, 2. Reaching close enough (ε)

Formula: $\boxed{x_{k+1} = x_k - \dfrac{f(x_k)}{f'(x_k)}}$

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Example: Given $f(x) = x^2 - 2xe^{-x} + e^{-2x}$, Initial point $x_0 = 1$ and Error bound $= 0.01$. Find root of $f(x)$ using Newton's method.

$\Rightarrow$ $f'(x) = 2x - [2e^{-x} + 2xe^{-x}(-1)] + e^{-2x}(-2)$

$= 2x - 2e^{-x} + 2xe^{-x} - 2e^{-2x}$

K	x_K	$f(x_K)$	$f(x_K) <$ Error Bound (0.01)	$f'(x_K)$	$x_{k+1} = x_K - \frac{f(x_n)}{f'(x_n)}$
0	1	0.3995	×	1.729	$x_1 = 0.7689$
1	0.7689	0.0933	×	0.8938	$x_2 = 0.6645$
2	0.6645	0.0225	×	0.4542	$x_3 = 0.6149$
3	0.6149	0.005506	✓		

Root, x^*

$\therefore x^* = 0.6149$

Example: $f(x) = \frac{1}{x} - 0.5$ [x^* is at 2], Given, initial point $x_0 = 1$, find the root of $f(x)$ with relative error at each iteration.

$\Rightarrow$ $f'(x) = \cancel{3}x - \frac{1}{x^2}$

k	x_k	$f(x_k)$	$f'(x_k)$	$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$	Relative error $= \lvert x_{k+1} - x_k \rvert$
0	1	0.5	-1	1.5	0.5
1	1.5	0.1667	-0.4444	1.875	0.375
2	1.875	0.0333	-0.284	1.9922	0.11725
3	1.99225	0.00194	-0.25	2.00	0.00775
4	2.00	0			

Root x^*

$\therefore x^* = 2$.

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** Important **

Prove that Newton Raphson Method is a super linear convergent.

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$\downarrow$$

$$g(x) = x - \frac{f(x)}{f'(x)}$$

We know,

Convergence Rate, $\lambda = |g'(x)|$

$$= \left| \frac{d}{dx}\left(x - \frac{f(x)}{f'(x)}\right) \right|$$

$$= \left| 1 - \frac{f'(x)\,f'(x) - f(x)\,f''(x)}{[f'(x)]^2} \right|$$

$$= \frac{[f'(x)]^2 - [f'(x)]^2 + f(x)\,f''(x)}{[f'(x)]^2}$$

$$\lambda = \left| \frac{f(x)\,f''(x)}{[f'(x)]^2} \right|$$

$$\lambda = |g'(x^*)| = \left| \frac{f(x^*)\,f''(x^*)}{[f'(x^*)]^2} \right|$$

$$= \left| \frac{0 * f''(0)}{[f'(0)]^2} \right|$$

$\lambda = 0 \longrightarrow$ Super Linear Convergent.

$\therefore$ Newton Raphson Method is super Linear convergent.

[Proved]

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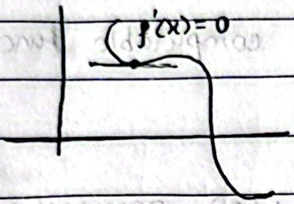
Advantage:

1. Since it is super linear convergent this approach is much faster.

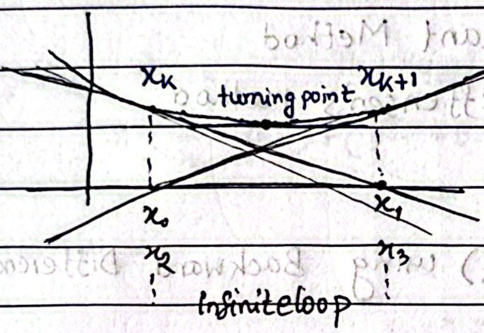
Disadvantage:

1. formula, $x_{k+1} = x_k - \frac{f(x)}{f'(x)} \longleftarrow 0$



$\longrightarrow$ if the iteration leads to a turning point ($f'(x)=0$) then we can't find the root as x_{k+1} becomes infinity.

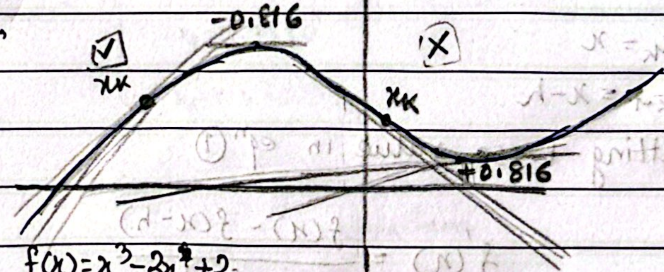
2.



If the initial point is chosen such that it lies near a turning point, the iteration may enter a loop and making it impossible to find the root.

Does not work if there is a turning point between x_k and x_{k+1}

let,



$f(x) = x^3 - 2x + 2$

Turning Point, $f'(x) = 3x^2 - 2 = 0$

$\Rightarrow x = \pm 0.816$

We need to make sure $x_0 < -0.816$

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4.7 : ~~Quassi~~ Quasi Newton Method / Second form :

From "Newton Raphson Method", $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$ → Two different function

* Computation cost is higher when we are calculating two different function.

So, we will replace $f'(x_k)$ by computable function g_k

$$f'(x_k) \approx g_k$$

This can be done in different (2) approach:

1. Secant Method
2. Steffensen's method.

⇒ Secant Method :

↳ Replaces $f'(x_k)$ using Backward Difference formula.

- **Derivation :**

We know,

Backward difference, $f'(x) = \frac{f(x) - f(x-h)}{h}$ ---- (1)

let. $x_k = x$

$x_{k-1} = x - h$

Putting these value in eqⁿ (1)

$$f'(x) = \frac{f(x) - f(x-h)}{x - (x-h)}$$

$$\Rightarrow f'(x_k) = \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}} \quad \text{——— (ii)}$$

Newton Raphson Formula,

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$= x_k - \frac{f(x_k)}{\frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}}$$

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$$\Rightarrow \boxed{x_{k+1} = x_k - \frac{f(x_k)\,(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})}}$$

Formula of secant method/Quasi-Newton Method.

Example: Given, $f(x) = \frac{1}{x} - 0.5$, find the root of $f(x)$ using ~~Qusi~~ Quasi Newton Method/Secant Method given initial point $x_1 = 0.5$, $x_0 = 0.25$.

$\Rightarrow$

$$x_{k+1} = x_k - \frac{f(x_k)\,(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})}$$

$$\Rightarrow x_{k+1} = x_k - \frac{\left(\frac{1}{x_k} - 0.5\right)(x_k - x_{k-1})}{\left(\frac{1}{x_k} - 0.5\right) - \left(\frac{1}{x_{k-1}} - 0.5\right)}$$

iteration, k	x_k
0	$x_0 = 0.25$
1	$x_1 = 0.5$
2	$x_2 = 0.6875$
3	$x_3 = 1.01562$
4	$x_4 = 1.3540$
5	$x_5 = 1.68205$
⋮	⋮
12	$x_{12} = 2.00$
13	$x_{13} = 2.00$

same

$x^* = 2.00$